Bellabeat Smart Device Data Analysis Case Study

1. Ask (Business Task)

Bellabeat is a wellness technology company that develops smart devices designed to help women track and improve their health and lifestyle. The company collects data from its smart products to help users understand their daily activity, sleep habits, and overall wellness.

The objective of this analysis is to explore smart device usage data to identify trends in user behavior. The insights generated from this analysis can help Bellabeat improve its marketing strategy and increase user engagement with its products.

Business Questions

1. What are the trends in smart device usage?
2. How can these trends apply to Bellabeat customers?
3. How can these trends influence Bellabeat's marketing strategy?

Key Stakeholders

- Bellabeat Executive Team
- Bellabeat Marketing Analytics Team
- Bellabeat Customers and Potential Users

The results of this analysis will help the company better understand user behavior and guide marketing decisions.

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2. Prepare (Data Source)

Dataset Description

The dataset used for this analysis is the **FitBit Fitness Tracker Data**, which is publicly available on Kaggle. This dataset contains personal fitness tracker data from 30 Fitbit users who consented to share their information.

The dataset includes information related to:

- Daily step counts
- Calories burned
- Activity intensity levels
- Sleep tracking data

Files Used in the Analysis

The following datasets were used:

- dailyActivity_merged.csv
- sleepDay_merged.csv

These datasets provide useful information about daily activity patterns, physical movement, and sleep behavior.

Data Organization

The data is stored in multiple CSV files where each file represents different health and activity metrics recorded by Fitbit devices.

Data Limitations

There are several limitations to consider:

- The dataset includes only 30 users, which is a relatively small sample size.
- The data was collected in 2016 and may not reflect current user behavior.
- The dataset is not specifically focused on women, even though Bellabeat's target audience is primarily women.

Despite these limitations, the dataset still provides valuable insights into smart device usage patterns.

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3. Process (Data Cleaning and Preparation)

Before conducting the analysis, the dataset was cleaned and prepared to ensure accuracy and consistency.

Tools Used

The following tools were used during the data preparation and analysis process:

- Google Sheets
- SQL (MySQL Workbench)
- R Studio

Data Cleaning Steps

The following steps were performed during the cleaning process:

1. Imported CSV files into Google Sheets.
2. Checked the dataset for duplicate records and removed duplicates where necessary.
3. Examined the dataset for missing or incomplete values.
4. Converted the date column into a proper date format.
5. Created a new column called **DayOfWeek** to analyze activity by day.
6. Ensured numeric columns were correctly formatted for calculations.
7. Exported the cleaned dataset for further analysis using SQL and R.

These steps helped ensure that the dataset was ready for accurate analysis.

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4. Analyze (SQL Data Analysis)

SQL was used to explore the dataset and calculate summary statistics. The cleaned dataset was imported into a SQL database and several queries were executed to identify patterns in user activity.

4.1 Total Number of Users

```
SELECT COUNT(DISTINCT Id)
FROM daily_activity_clean;
```

COUNT(DISTINCT Id)
7

Result 3

4.2 Average Daily Steps

```
SELECT AVG(TotalSteps)
FROM daily_activity_clean;
```

AVG(TotalSteps)
6507.4722

Result 4

4.3 Maximum and Minimum Steps

```
SELECT MAX(TotalSteps), MIN(TotalSteps)
FROM daily_activity_clean;
```

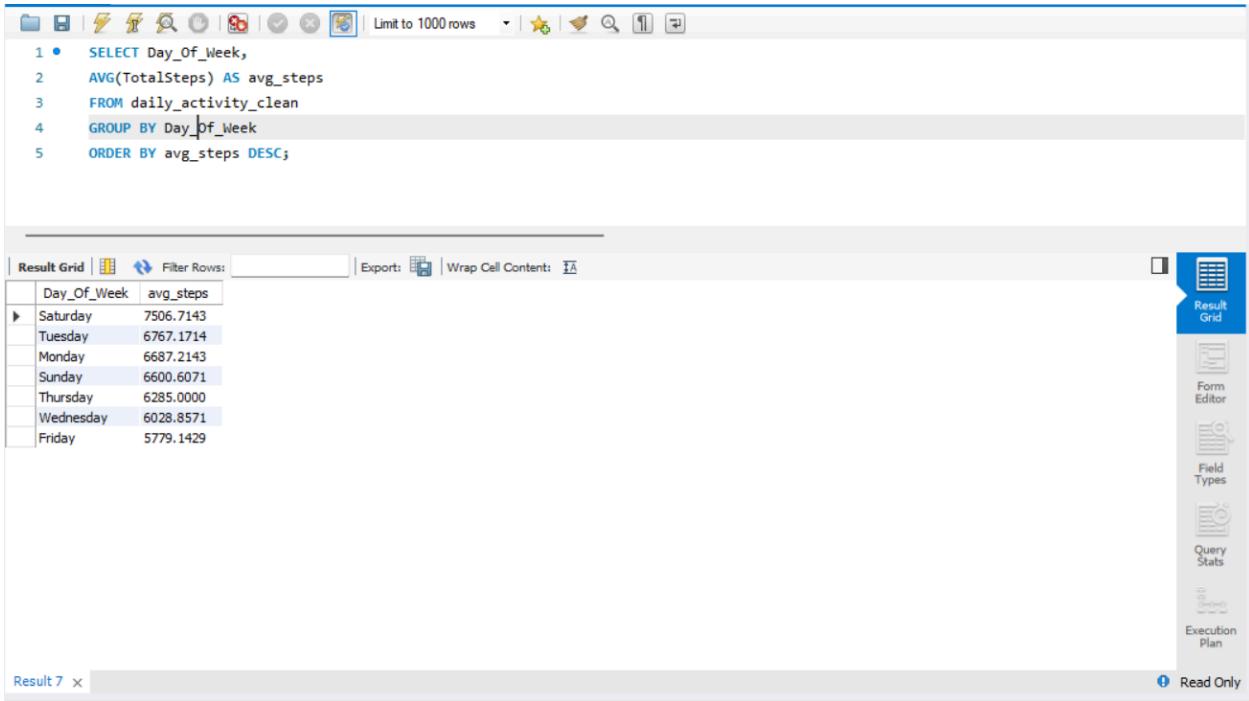
MAX(TotalSteps)	MIN(TotalSteps)
36019	0

4.4 Average Calories Burned

```
SELECT AVG(Calories)
FROM daily_activity_clean;
```

AVG(Calories)
1983.0370

4.5 Average Steps by Day of Week



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5. Share (Data Visualization)

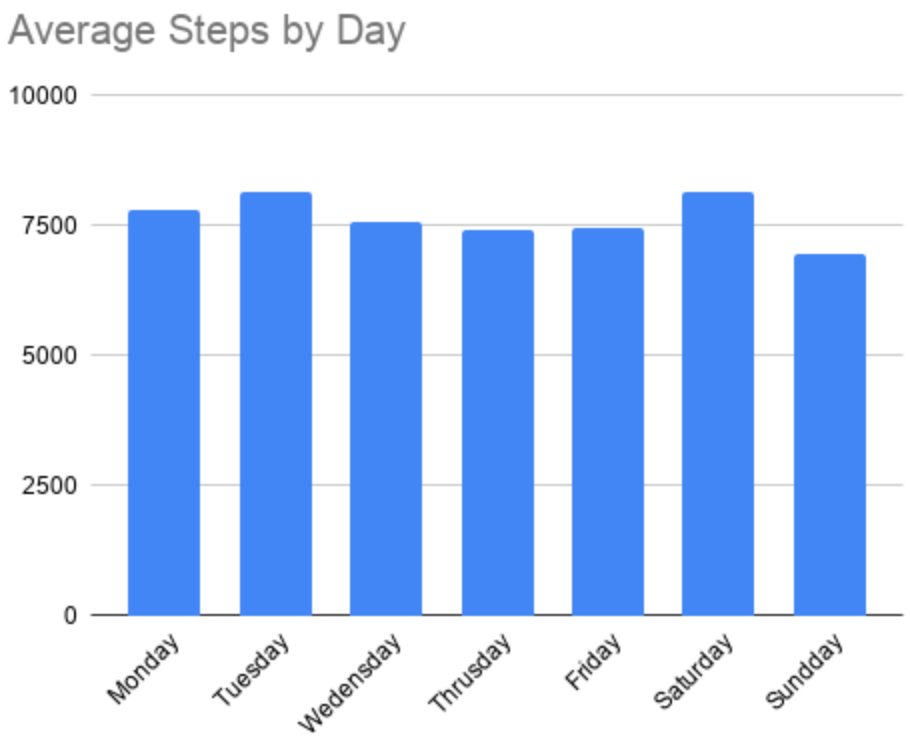
To present the results clearly, several visualizations were created using Google Sheets.

Visualizations Created

1. Average Steps by Day

Chart Type: Column Chart

This chart shows how the average number of steps changes throughout the week.



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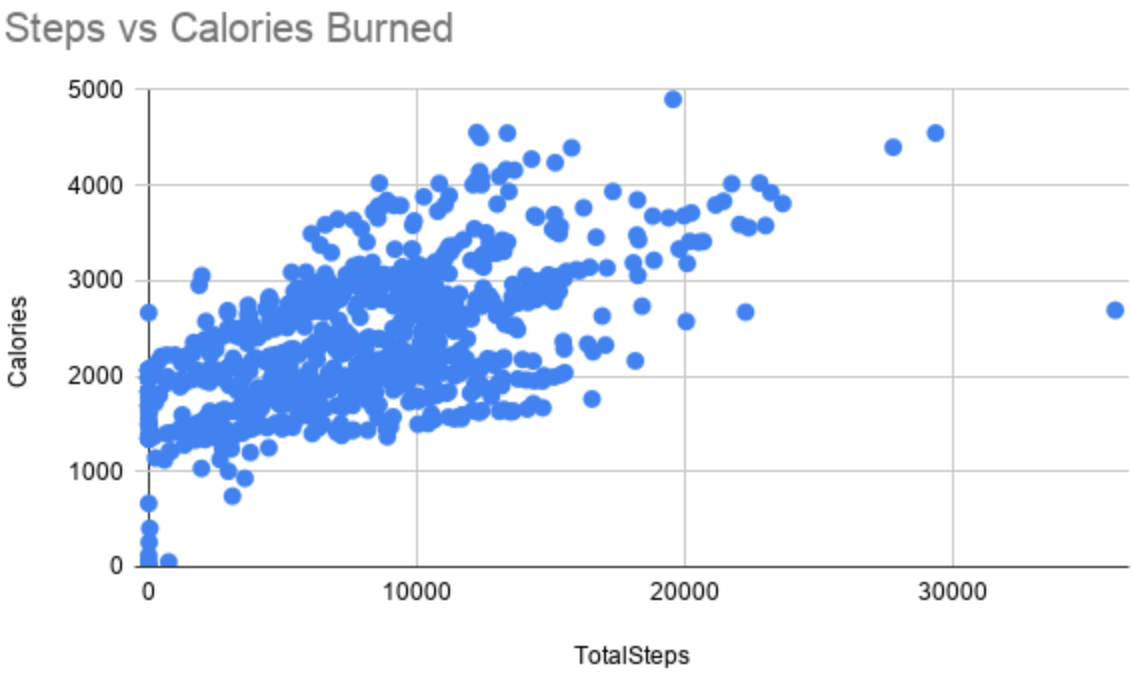
2. Steps vs Calories Burned

Chart Type: Scatter Plot

This visualization shows the relationship between the number of steps taken and the calories burned.



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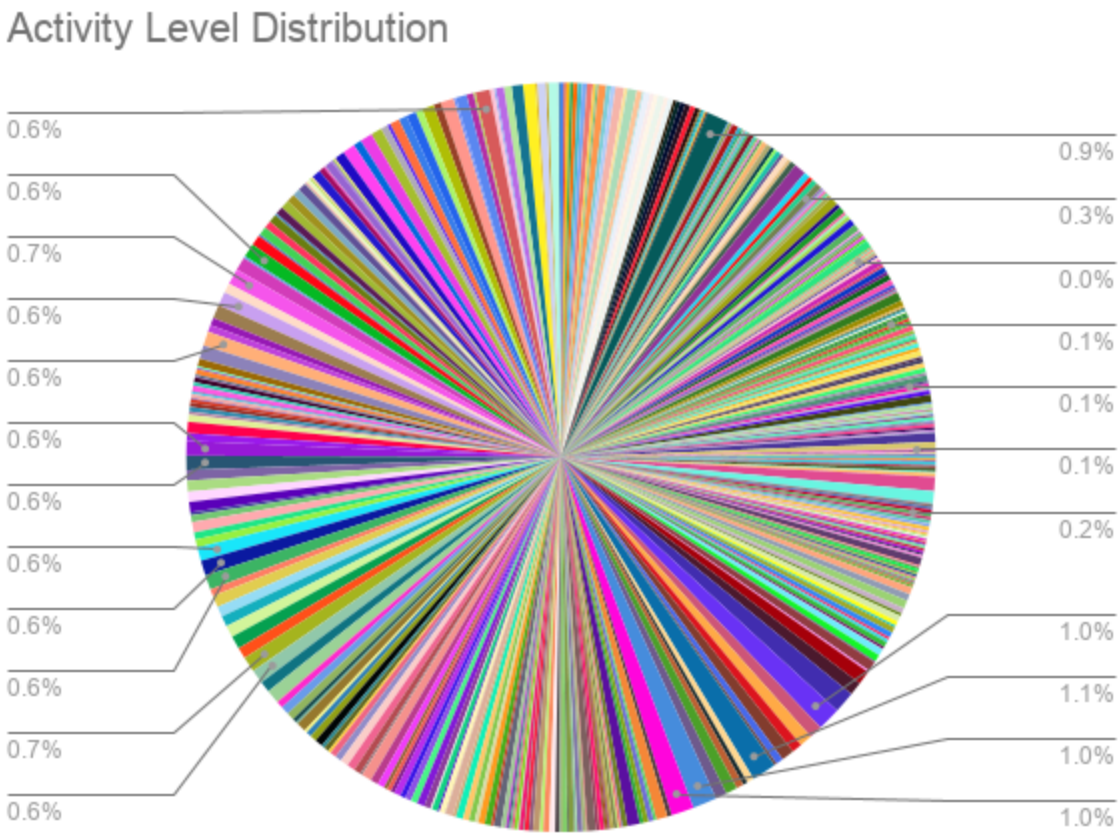
3. Activity Level Distribution

Chart Type: Pie Chart

This chart displays the proportion of time users spend in different activity levels, including:

- Very active
- Fairly active
- Lightly active
- Sedentary



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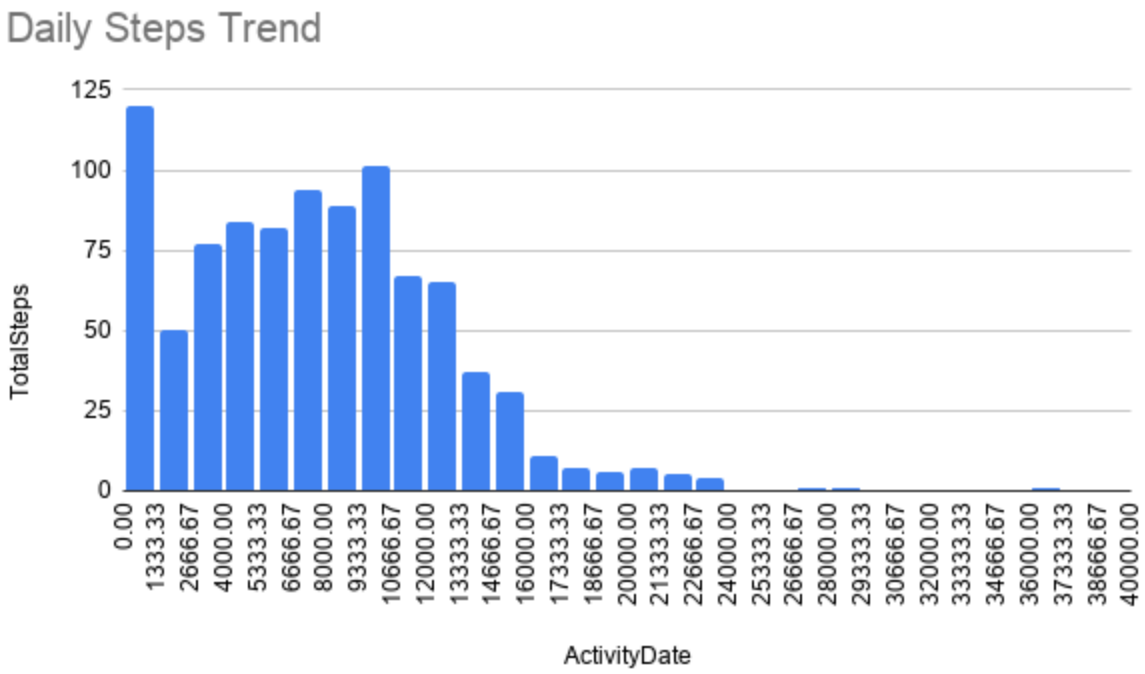
4. Daily Steps Trend

Chart Type: Histogram Chart

This chart shows how daily step counts fluctuate over time.

These visualizations help communicate insights clearly and support the analysis results.



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6. Act (Insights and Recommendations)

Based on the analysis, several key insights and recommendations were developed for Bellabeat.

Key Insights

1. There is a strong relationship between steps taken and calories burned.
2. Users tend to be more active during weekdays compared to weekends.
3. A large portion of user time is spent in sedentary or lightly active states.
4. Daily activity levels vary significantly across different days.

Recommendations

1. Encourage Daily Activity

Bellabeat can motivate users by setting daily step goals and sending reminders through the Bellabeat mobile application.

2. Promote Weekend Fitness

Since activity levels appear lower during weekends, Bellabeat can introduce weekend fitness challenges or reward systems.

3. Provide Personalized Health Insights

Bellabeat can use user data to provide personalized health and wellness recommendations to help users improve their habits.

Conclusion

This analysis of smart device fitness data provided valuable insights into user activity patterns and health behaviors. By understanding how users interact with their fitness tracking devices, Bellabeat can develop more effective marketing strategies and improve user engagement.

Encouraging consistent activity, providing personalized insights, and promoting healthy habits can help Bellabeat strengthen its position in the smart wellness device market.